SOFTWARE REQUIREMENTS SPECIFICATION

CampusFix

Intelligent Campus Maintenance Prioritization & Resolution Management System

Software Engineering Project
Formal SRS Document

Document Item	Details
Version	1.1
Status	For Faculty Review
Project Type	Software Engineering
Prepared By	Project Team

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements of CampusFix, an Intelligent Campus Maintenance Prioritization & Resolution Management System. It establishes expected system behavior, interfaces, constraints, quality requirements, and operational rules.

1.2 Scope

CampusFix centralizes the reporting, prioritization, assignment, tracking, resolution, and verification of campus maintenance issues. Students and faculty report problems. Maintenance teams receive assignments, update progress, and publish maintenance activities. The Main Admin supervises the workflow.

- Campus areas include washrooms, classrooms, laboratories, Wi-Fi/network, buses, parks, parking, hostels, and internal roads.
- Water complaints include unusual taste, color, smell, dirty water, low pressure, no supply, leakage, and suspected contamination.
- The system supports Critical, High, Medium, and Low priorities.
- Related complaints can be grouped into a common incident.
- Maintenance teams can publish ongoing or planned maintenance activities.
- The current academic scope focuses on software documentation; production website deployment is not required.

1.3 Definitions, Abbreviations

Term	Definition
SRS	Software Requirements Specification
CampusFix	Proposed campus maintenance management system
Complaint	Maintenance issue submitted by an authorized user
Incident	Group of related complaints representing one underlying issue
SLA	Target response or resolution time
Admin	Main administrator
Maintenance Team	Team responsible for assigned maintenance work
UML	Unified Modeling Language
DFD	Data Flow Diagram
FR	Functional Requirement
NFR	Non-Functional Requirement

1.4 References

- Faculty-provided SRS structure.
- CampusFix requirements proposed for faculty approval.
- Software Engineering course material on requirements engineering.
- UML and DFD conventions used in the course.
- Project Jira and GitHub documentation.

1.5 Overview

Section 1 introduces the project. Section 2 describes the system, users, constraints, assumptions, and process conditions. Section 3 defines functional, non-functional, interface, and data requirements. Section 4 describes external interfaces. Section 5 explains system features. Section 6 covers legal, ethical, maintenance, and support considerations.

2. Overall Description

2.1 Product Perspective

CampusFix is a centralized maintenance coordination system connecting campus users, maintenance teams, and the Main Admin. It converts individual reports into a workflow involving review, prioritization, assignment, resolution, verification, and closure.

2.2 Product Functionalities

- Role-based login for Student, Faculty, Maintenance Team, and Main Admin.
- Campus area and category selection.
- Issue reporting with location, description, severity, and optional photographs.
- Water-quality and supply reporting.
- Priority assignment using impact factors.
- Grouping duplicate or related complaints.
- Maintenance team assignment and progress tracking.
- SLA monitoring and escalation.
- Resolution verification, feedback, and reopening.
- Campus maintenance activity announcements.
- Preventive maintenance scheduling.
- Notifications and administrative reports.

2.3 User Characteristics

User	Characteristics and Responsibilities
Student	Reports issues, tracks complaints, receives alerts, and provides feedback.
Faculty	Reports and tracks issues and receives maintenance alerts.
Maintenance Team	Accepts assignments, performs maintenance, updates progress, records resolutions, and publishes activities.
Main Admin	Manages users/categories, reviews and prioritizes complaints, assigns teams, escalates issues, verifies resolutions, and views reports.

2.4 Constraints

- The system shall follow the faculty-approved requirements baseline.
- The project must remain feasible for an academic team.
- UML, DFD, Jira, and test artifacts must remain consistent with this SRS.
- Only authorized users may perform role-specific operations.
- Priority and SLA rules must be documented and approved.
- CampusFix reports water concerns but does not certify water safety.
- Production website deployment is outside the current academic scope.

Module-specific constraints:

Module	Constraint
Issue Reporting	Requires a valid user, campus area, and category.
Prioritization	Uses documented and approved factors.
Maintenance Assignment	Assignments should match team expertise.
Water Complaints	Suspected contamination requires authorized review.
Maintenance Activities	Posts require location and timing details.

Administration	Sensitive operations are restricted to the Main Admin.

2.5 Assumptions and Dependencies

- Users have valid credentials.
- Campus areas and categories are configured by the Main Admin.
- Maintenance teams are organized by expertise.
- Authorized personnel verify critical decisions and completed work.
- Notification delivery depends on the selected mechanism.
- SLA values require faculty/campus approval.
- The system requires persistent storage for records.

2.6 Preconditions and Postconditions

Process	Precondition	Postcondition
Submit complaint	User is authenticated and selects a valid category.	Unique complaint ID is generated; report enters review.
Assign issue	Complaint is reviewed and prioritized.	Suitable team receives assignment.
Resolve issue	Team has accepted the assignment.	Resolution details are recorded.
Verify resolution	Issue is marked resolved.	Issue is closed or reopened.
Publish activity	Maintenance user is authorized.	Activity becomes visible to relevant users.
Escalate issue	Critical or overdue criteria are met.	Responsible authority is notified.

3. Specific Requirements

3.1 Functional Requirements

The following 25 functional requirements define the core system behavior. Further sub-requirements may be added if approved by the faculty.

ID	Requirement
FR-01	The system shall provide login access for Students, Faculty, Maintenance Team members, and the Main Admin, with role-based permissions.
FR-02	The system shall allow authorized users to log out and shall prevent access to unauthorized functions.
FR-03	The system shall allow Students and Faculty to submit maintenance complaints.
FR-04	The system shall allow a reporter to select a campus area and the corresponding issue category.
FR-05	The system shall capture issue description, specific location, severity, submission time, and optional photographs.
FR-06	The system shall generate a unique identifier for every submitted complaint.
FR-07	The system shall support washrooms, classrooms, laboratories, Wi-Fi/network, transport, parks, parking, hostels, and internal roads.
FR-08	The system shall support water complaints involving taste, color, smell, dirty water, low pressure, no supply, leakage, or suspected contamination.
FR-09	The system shall flag suspected water-contamination complaints for urgent review by authorized personnel.

FR-10	The system shall assign Critical, High, Medium, or Low priority using approved factors such as severity, safety impact, affected users, location importance, waiting time, and repeated reports.
FR-11	The system shall identify potentially related complaints and allow authorized users to group them into one incident.
FR-12	The system shall track the number of related reports or affected users for a grouped incident.
FR-13	The system shall assign complaints to maintenance teams based on issue category and team expertise.
FR-14	Maintenance Team members shall view assignments, accept work, and update progress.
FR-15	Maintenance Team members shall record resolution details and attach supporting evidence where required.
FR-16	The system shall track complaint status from submission through review, prioritization, assignment, work, resolution, verification, and closure.
FR-17	Authorized personnel shall verify a resolution, and unresolved or recurring issues shall be eligible for reopening.
FR-18	The system shall support configurable response or resolution targets, identify overdue complaints, and escalate critical or overdue issues.
FR-19	The system shall notify relevant users about complaint status changes, assignments, escalations, and resolution outcomes.
FR-20	Maintenance Team members shall publish planned or ongoing maintenance activities with title, location, start time, expected completion, and affected area.
FR-21	The system shall display active maintenance activities and support alternative-route or alternative-location information where applicable.
FR-22	The system shall support preventive maintenance schedules, reminders, completion records, and maintenance history.
FR-23	The system shall provide the Main Admin with a dashboard showing complaint counts, priorities, categories, locations, resolution times, and recurring issues.
FR-24	The system shall maintain records of complaints, assignments, resolutions, maintenance activities, notifications, and preventive maintenance.
FR-25	The system shall allow users to provide feedback after resolution and retain that feedback with the complaint record.

3.2 Non-Functional Requirements

ID	Requirement
NFR-01	The system shall enforce role-based authorization for protected functions.
NFR-02	The system shall protect user credentials and restrict access to personal information.
NFR-03	The system shall validate uploaded files and protect stored photographs and evidence.
NFR-04	The interface shall provide clear navigation and understandable messages.

NFR-05	The issue-reporting workflow shall be straightforward for regular campus users.
NFR-06	The system should respond within an acceptable time under normal expected usage; measurable targets shall be set during design.
NFR-07	The system should support multiple users working concurrently.
NFR-08	The system shall preserve complaint records and maintain consistent status history.
NFR-09	The system should minimize accidental data loss through appropriate storage and backup practices.
NFR-10	The system shall be maintainable through configurable categories, priorities, and SLA rules.
NFR-11	The system should support future enhancements without requiring a complete redesign.
NFR-12	The system shall provide reliable access to authorized users when the service is available.

3.3 Interface Requirements

- The interface shall provide role-specific dashboards.
- Issue reporting shall follow a clear area → category → details flow.
- Complaint status and priority shall be clearly displayed.
- Maintenance alerts shall show affected locations and expected duration.
- Admin views shall support filtering and summary information.
- The interface shall validate required fields and display understandable error messages.

3.4 Data Requirements

Data requirements are organized into the three parts specified in the course format: data storage, data retrieval, and data formats.

3.4.1 Data Storage

- Store user profiles and role information.
- Store campus areas and issue categories.
- Store complaints, incident groupings, priorities, and status history.
- Store assignments, resolution details, and verification records.
- Store maintenance activity announcements and preventive maintenance schedules.
- Store notification records and resolution feedback.
- Protect personal information and restrict records according to user role.

3.4.2 Data Retrieval

- Users shall retrieve their complaint details and status updates.
- Maintenance teams shall retrieve assigned complaints and relevant activity details.
- The Main Admin shall retrieve complaints by status, priority, category, location, and date.
- Authorized users shall view active maintenance activities and expected completion times.
- The system shall retrieve historical complaint, resolution, and preventive maintenance records for reporting.

3.4.3 Data Formats

- Each complaint shall have a unique identifier.
- Dates and times shall use a consistent system-wide format.
- Priority and status shall use predefined values.

- Descriptions and comments shall be stored as text.
- Photographs and evidence shall use approved image/file formats and size limits.
- Required fields shall be validated before a record is saved.

4. External Interface Requirement

4.1 Hardware Requirement

- Standard computer or mobile device with display and input capabilities.
- Optional camera for issue photographs.
- A server or hosting environment if the system is deployed.

4.2 Software Requirement

- Application runtime environment selected during design.
- Database management system for persistent records.
- Authentication and authorization mechanism.
- File or media storage for photographs and evidence.
- Notification mechanism if included in implementation.
- GitHub for version control of documentation and project artifacts.

4.3 Communication Interface

- Secure communication between the user interface and application.
- Communication between application components and the database.
- Communication with notification services where applicable.
- Network connectivity for accessing the system.

5. System Features

Feature	Description
5.1 Role-Based Access	Provides separate permissions for Student, Faculty, Maintenance Team, and Main Admin.
5.2 Issue Reporting	Captures campus area, category, description, location, severity, and optional photographs.
5.3 Water Quality and Supply	Supports reports about unusual taste, color, smell, dirty water, pressure, supply interruption, leakage, and suspected contamination.
5.4 Prioritization	Uses approved impact factors to assign a priority level.
5.5 Related Complaint Grouping	Groups related reports to reduce duplicate maintenance assignments.
5.6 Team Assignment	Routes issues to suitable Electrical, Plumbing, Civil, IT/Network, Housekeeping, Security, or Transport teams.
5.7 Resolution Tracking	Tracks work progress, resolution, verification, feedback, and reopening.
5.8 SLA and Escalation	Monitors target times and escalates critical or overdue issues.
5.9 Campus Maintenance Alerts	Publishes ongoing or planned work with locations, time windows, and optional alternative routes.
5.10 Preventive Maintenance	Supports schedules, reminders, completion records, and history.
5.11 Notifications	Provides complaint, assignment, escalation, and activity updates.
5.12 Admin Reporting	Summarizes complaints, priorities, locations, categories,

	resolution times, and recurring issues.

6. Other Requirements

6.1 Legal and Regulations

- The system shall follow applicable institutional policies for data handling.
- Personal information shall be collected only where necessary.
- Uploaded photographs and user information shall be handled according to applicable privacy requirements.
- Administrative access shall be limited to authorized personnel.
- The system shall not claim to certify water safety or replace laboratory testing.

6.2 Ethical Considerations

- Protect user privacy and avoid unnecessary exposure of reporter identity.
- Use anonymous reporting responsibly for approved cases.
- Apply documented priority rules consistently.
- Do not manipulate complaint records or priority decisions unfairly.
- Provide truthful complaint and maintenance status information.
- Publish accurate activity locations and time windows to help users avoid affected areas.

6.3 Maintainability and Support

- Maintain requirements and project documents under version control in GitHub.
- Allow authorized administrators to update categories and approved rules.
- Keep complaint and maintenance histories available for review.
- Document changes to requirements and related diagrams/test cases.
- Use consistent identifiers across the SRS, UML, DFD, Jira, and tests.

Software Evolution

1. Identify a requested requirement change.
2. Review the change with the project team and faculty where necessary.
3. Update and version the requirement document.
4. Identify affected UML diagrams, DFDs, Jira items, and test cases.
5. Update affected artifacts and review consistency.
6. Approve and retain the next baseline version.

Appendix A — Proposed Complaint Lifecycle

Reported → Reviewed → Prioritized → Assigned → Accepted → In Progress → Resolved → Verified → Closed

Appendix B — Faculty Review and Approval

Review Area	Faculty Comments / Decision
Scope and definitions	
Overall description and constraints	
Functional requirements (25)	
Non-functional requirements	

Three data requirement parts	
External interfaces and system features	
Other requirements	
Final status	☐ Approved ☐ Changes Required

Faculty Name	**Signature**	**Date**